

Debugging the Body

A Personal Journey to Sustainable Health in Tech

Kyle Jenkins

Principal Consultant w/ [improving](#) 

Who am I?

20 Years in Tech

Video Gamer

Public Speaking 10 Years

Crochet

User Group Supporter

Health nut in my own way

Agenda

Why / Goals

Caveats / Thoughts

My Story

My Health Manifesto

Details – The Research

My Personal Strategies – Putting it Together

And then what?

Why

Family



Health



Friends



Work



Why



Why

~\$8000 Preventable Costs

Insurance Sucks

Myths – AI is not helping

Take care of it now - it will only get worse

Vulnerability Creates Conversation

Help YOU define your journey

Caveats / Thoughts

My Solutions may not work
for you – Results Overlap

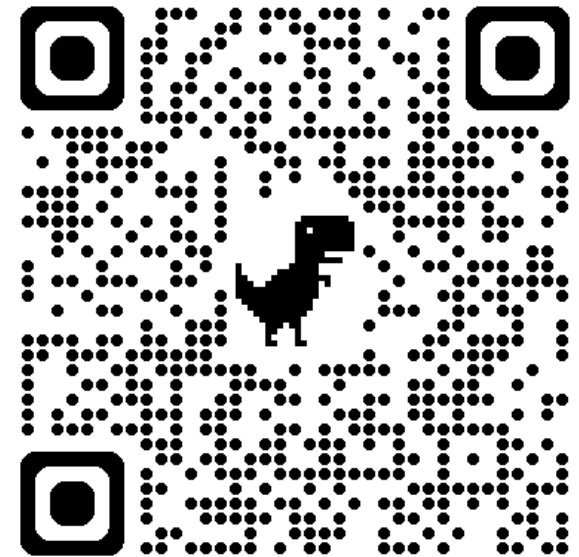
Choice & Experimentation

There is Black, White, and
Grey

Whatever you plan: Review
with your doctor

For the Women:

<https://www.youtube.com/watch?v=kizDk8idpT8>



My Story

My Story – Columbus and Fear - 2016



A1C

Cholesterol / LDL

My Story – Columbus and Fear - 2016

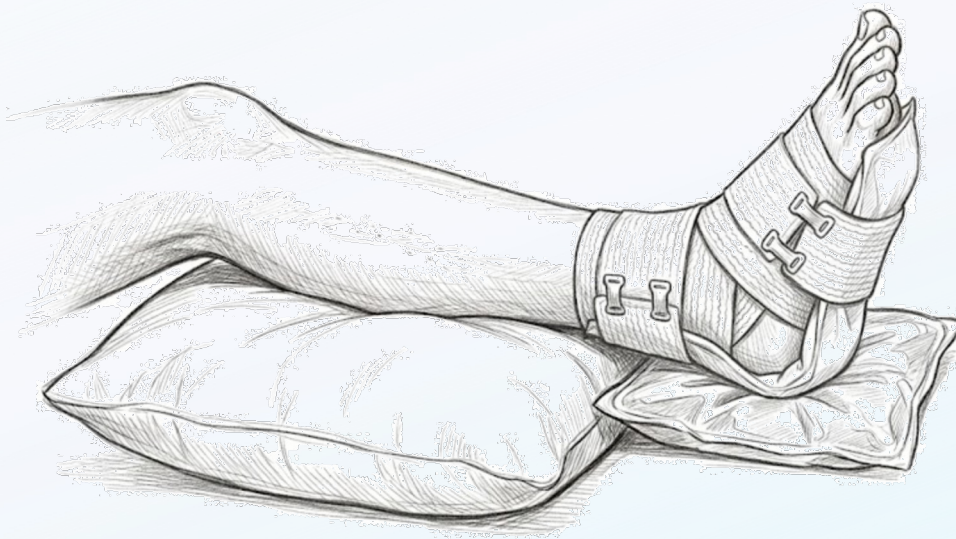
Dramatically reduce
sugar intake

Bought a Bike – Yay
Netflix!



My Story – Foot Pain - 2017

Pantar Fasciitis 1 of 3



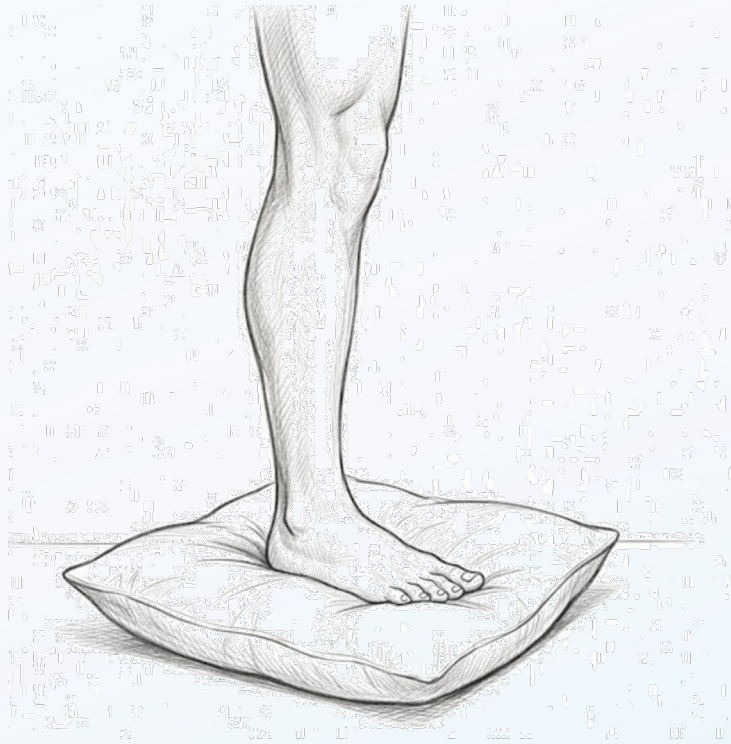
My Story – Building Habits - 2018

Struggling to get in the
bike: Introduced to
“Mini Habits”

60 Seconds on the Bike
Every day



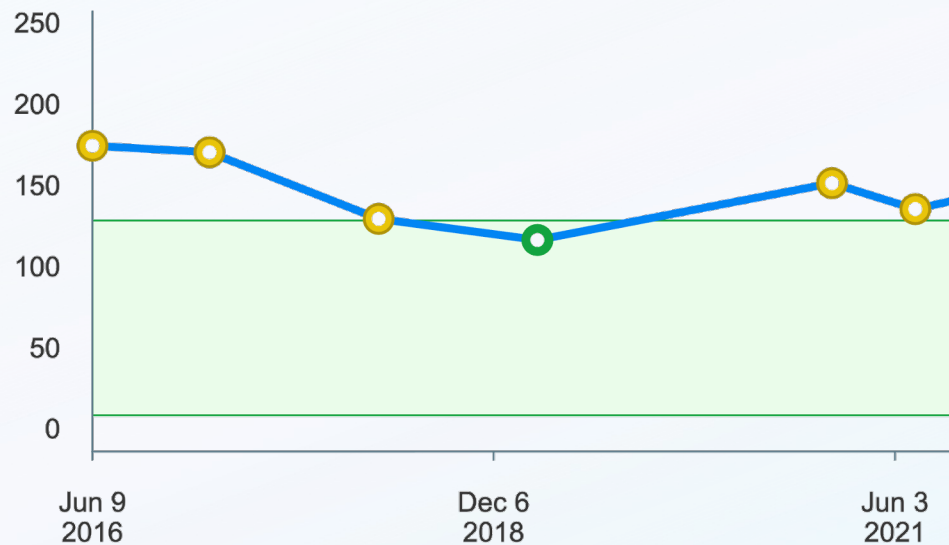
My Story – Foot/Knee Pain - 2019



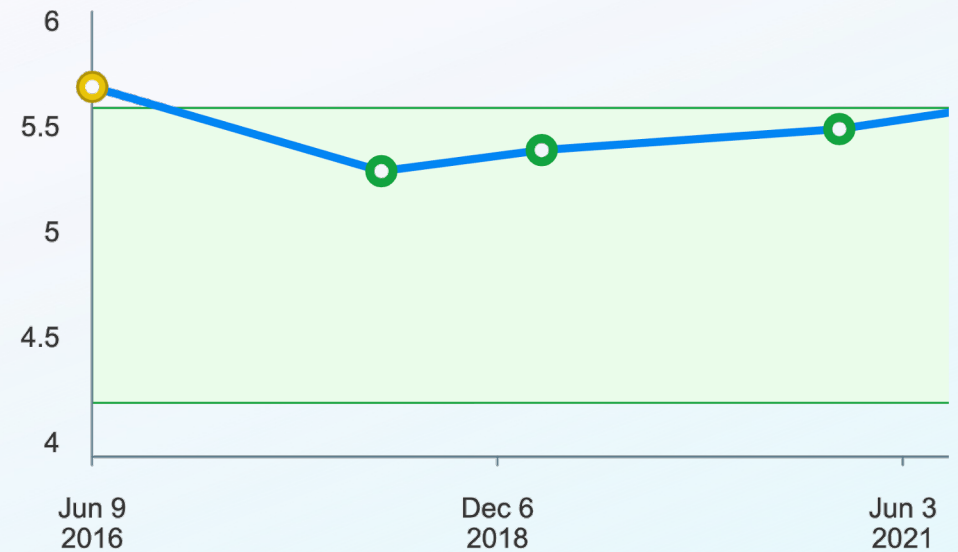
Right Knee & Foot Physical Therapy

My Story – Through 2021

LDL Calculated
mg/dL



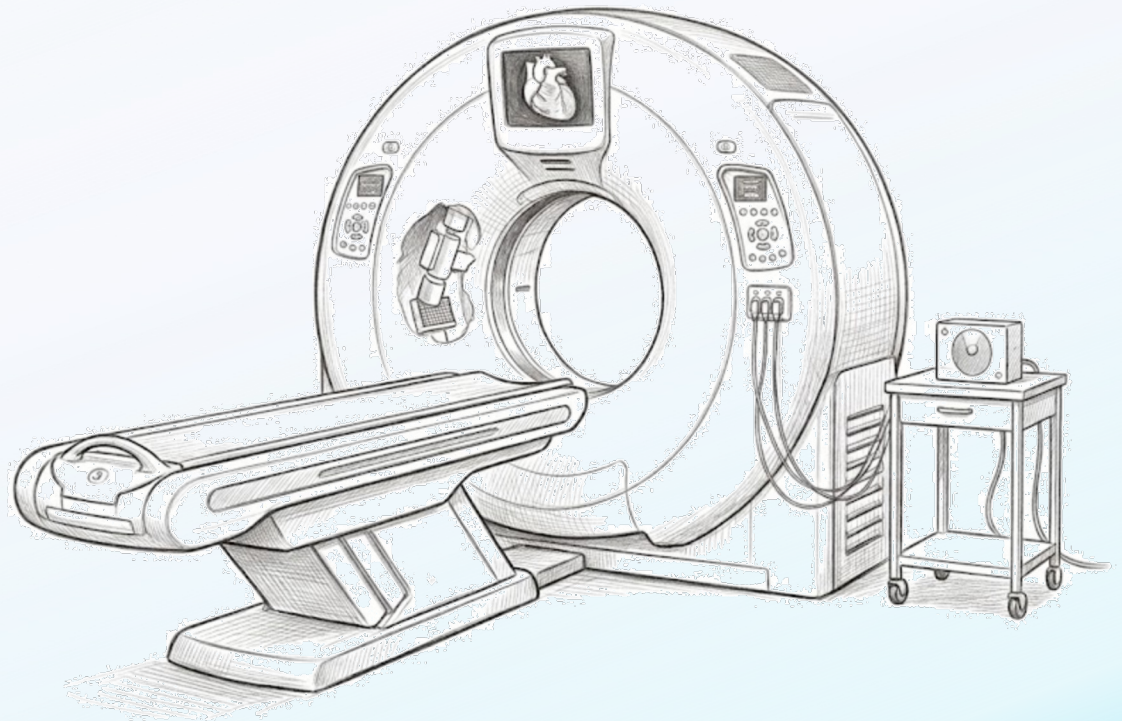
Hemoglobin A1C
Normal value: <5.7 %



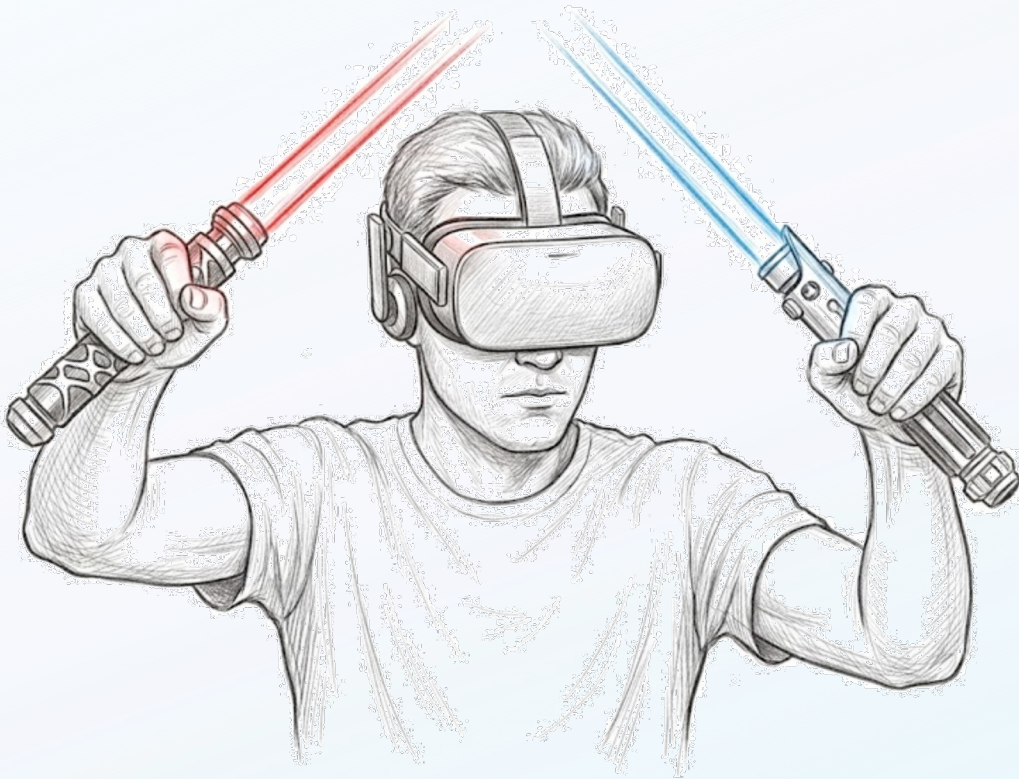
My Story – Doctor Shift - 2023

More Measurements

- CT Scoring: 0!
- % Body Fat
- Grip Strength
- Apolipoprotein B (ApoB)
134 (<90 ideal)
- LDL-P
2042 (<935 ideal)



My Story – New Plan - 2023

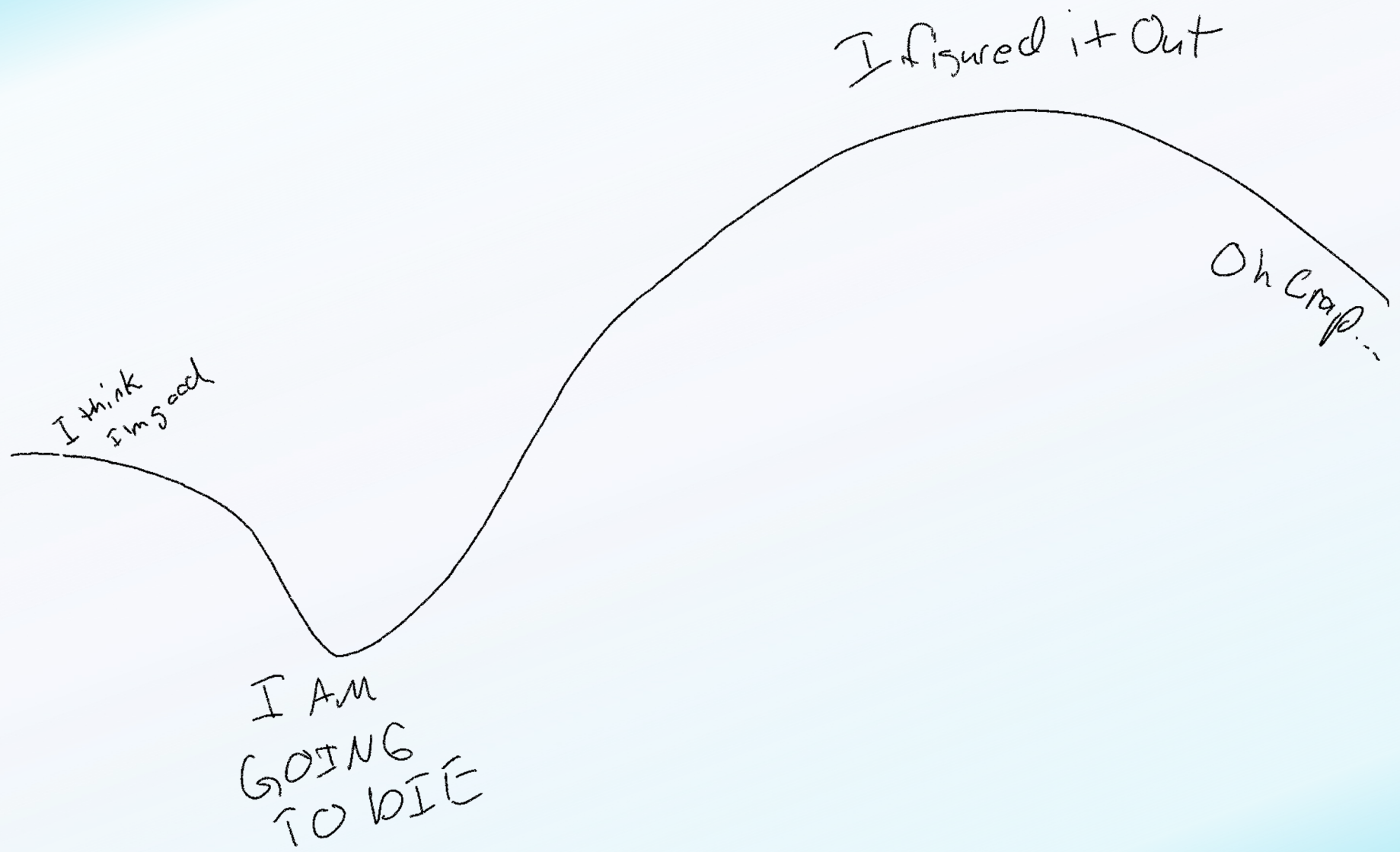


VR Headset – Fun +
Cardio

Building Lean Muscle

70g Protein

25g Fiber

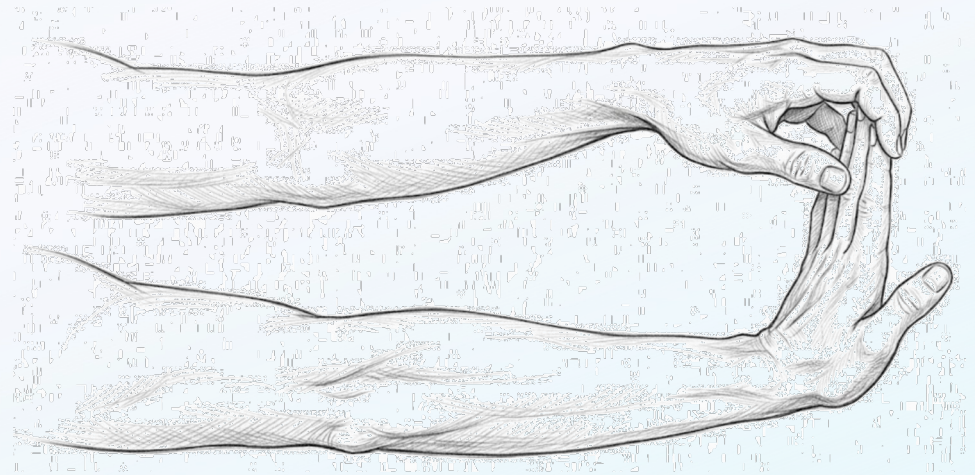


Numb Fingers

Diagnosis #1: Cubital Tunnel (Golfer's Elbow)

Treatment:

- Physical Therapy
- Night Splints
- Compression



Things got worse

Diagnosis #2: Carpal Tunnel

- Redirected to specialist... 3 weeks away
- One major oddity: my grip strength went up!
- For me, surgery is last resort

My strategy: Extreme Rest w/ Strength Building



Hey Dr Rubenstein. I'm looking forward to the appointment on the 9th. In the mean time, I can't do nothing while waiting, so I am going to do what I can to lay out "Here it is" and "Here is what I'm doing until we meet". If you think I am doing anything hazardous, please tell me ASAP, otherwise I will continue routine/process until we meet.

Key note: I have to have a specific routine, completely clear, no room for leeway. Otherwise, my brain cannot process, I question what I am doing, and things fall apart.

Timeline: Ulnar Nerve Diagnosis (Sept, 18th) -> PT @ NovaCare (Weekly) -> Increased range of numbness and symptoms -> Median Nerve diagnosis (Dec. 13th) -> Current Routine until we meet.

Issue list:

- Numbing/tingling of fingers during activities, major being: Sleeping, Driving, specific PT activities. Fingers effected: Thumb through ring finger, both hands. Current Diagnosis*, median nerve/carpel tunnel.
 - Paying attention, left fingers tingling comes from lifting left arm, similar position to driving.
- Pain in Right Elbow area. Diagnosis* of golfers elbow [ulnar nerve] (original symptoms included right/middle finger numbness, other fingers afterwards). Primarily while driving, cooking, VR, compression sleeves reduces pain during driving. *Unknown which pressure points effected (wrists/elbow/shoulder/neck).*
- [No diagnosis*] "Vibration" in hands moving fists up/down or gripping something solid. Assuming median nerve/carpel tunnel.
- ([#4])New, no diagnosis*] Pain/Weakness in Right Pinky/Ring MCP Joints. (I have a theory, will talk when I see you)
- [New, no diagnosis*] Left wrist pain washing dishes (dish holder). Assuming Carpel Tunnel.
- [No diagnosis*] Thumb pain near MCP (on/off). Assuming potential repeat of **DeQuervains**.
 - If pinched below left thumb MCP, there is noticeable pain.
- [No diagnosis*] Leaning on elbows may quickly create tingling/reduced feeling in fingers (Thumb through middle)

*diagnosis given by a doctor based on my understanding at the time. If "No diagnosis" I have no professional doctor statement. Assumptions noted are self driven.

Current Routine: Concept(s):

- Rest [splinting], Stretch [PT] & Strengthen [Muscle Building]. (Prednisone Completed 12/25)
- If I have to drop weights, that tells me I'm atrophying and may reduce splinting time / increase strength reps

Rest: Splinting wrists for near 23 hours per day (minus standing around networking at holiday parties). Other routines during the other 1 hour each night. This is because I'm always using my wrists (lifting/driving/typing/gaming/etc.), so taking them on/off is unrealistic, and wearing them while sleeping. Splinting right elbow @ 135 degrees during sleep, considering increasing that to as often as possible while at home/at computer [I'm just rarely at home] due to increased consistency in pain and no idea what else to do.

PT/Strength: 1 Hour each night, splints off				PT/Strength: Throughout day, Splints on [targeting Ulnar nerve], both arms/shoulders and neck:			
• Wrist Flexor stretch (2x 30 seconds) (Palm Up) • Wrist Flexor stretch (2x 30 seconds) (Palm Down) • Wrist Flexion w/ Dumbbell (3lbs x 10 reps) [Causes Vibrations; immediately ends at end-of-exercise] Removed due to fingers numbing • Wrist Grip Strengthen (~50-55kg), 4 sets of 10, each hand. [Started feeling weakness issue #4 here, can still do it though] • 1/1: Weakness issue feels gone • Rubber Band Outer Finger Pull (6.6lbs), 4 sets of 10, each hand. [Started feeling weakness issue #4 here, can still do it though] • 1/1: Weakness issue feels gone • Resistance Bars (2x 15) [moved from splints on section] • Daily Strength: • Push Ups on Knees (3 x 10) • Rotating Crunches (2 x 10), 15lbs • Mix (3-4 per day) of 2 sets of 10 of a variety (Caliber Routine), rotating:				• Forearm Supination/Pronation (5lbs x 10 reps) [Causes Vibrations; immediately ends at end-of-exercise] Removed due to fingers numbing • Resistance Bars: Ns, Us, Twists (2x 10-15 each) • Resistance band rows/pull down (2x 10 each) • Resistance band Ws (and whatever the face down is) (2x 10 each) • Chin Tuck (2 x 10 x 3 seconds) • Scalene/Upper Trap Stretch (2x 20 seconds) • Bilateral Pec Stretch in Doorway (2 x 20 seconds) • Nerve Glides (2 x 10) Removed due to fingers numbing			
DB Floor Press (35lbs)	DB Deadlift (30lbs)	DB Standing Shoulder Press (30lbs)	DB Goblet Squats (20lbs)				
DB Floor Fly (35lbs)	DB Renegade Row (35lbs)	DB Lateral Raise (15lbs)	DB Lunge (15lbs)				
DB Standing Tricep Extension (35lbs)	DB Standing Bent-Over Row (30lbs)	DB Standing Rear Lateral Raise (15lbs)	DB Bulgarian Split Squat (15lbs)				
	DB Curl (20lbs)		DB Romanian Deadlift (25lbs)				
DB = Dumbbell Note: No exercises noted involved bending wrists. (12/25: reducing to 1 set per for a round through) (12/31: 2 sets [1]) (1/1: 2 sets [2])							
Also shower and wash dishes during this time.							

Additional notes:

- Since starting routine, not having numbness waking up. I think the heavy splinting of wrists is actually helping here, first thing to help in months (before I was more focusing elbows, until learning about Median nerve last Friday).
- Right elbow pain is being more consistent, not just during activities. I'm more worried about this because I'm on the steroid and it's not reducing this.
- I would use a thumb spica splint instead, especially while sleeping, however I cannot find one that does not actively injure my skin. I had a custom (thermoplastic) one made previously for my left wrist/thumb which I could wear long term, but I accidently melted it in the back seat of my car, and have no idea how to get it replaced.

My Self full diagnosis, based on severity of symptoms:

- Carpal tunnel (Both Hands) (High)
- Right Arm Ulnar Nerve (High)
 - Possible Pronator Syndrome (Median Nerve)?
- Left Hand **DeQuervains** (Moderate)
- Left Arm Ulnar Nerve (Minor/Moderate)
 - Possible Pronator Syndrome (Median Nerve)?
- Right Hand **DeQuervains** (Minor)



Diagnosis #2: Carpal Tunnel – Test Results

No Sign of Carpal Tunnel

No Sign of Cubital Tunnel

Some irritation coming from the neck...

Diagnosis #3: Tech Neck



Chat GPT Summary of Results. Check with Doctor for accuracy / additional readings.

Key Findings:

1. **C6-C7 (Lower Neck):** There is a buildup of bone spurs and disc changes that are causing some narrowing in the center of the spinal canal (where the spinal cord runs). The right side is more affected, with a significant narrowing of the space where nerves exit the spine.
2. **C5-C6 (Mid-Lower Neck):** Similar bone and disc changes are present, causing mild-to-moderate narrowing in the central spinal canal and moderate narrowing of the areas where nerves exit on both sides.
3. **C4-C5 (Mid-Neck):** Bone and disc changes are worse on the left side, leading to mild narrowing in the central spinal canal and moderate-to-severe narrowing of the nerve passageways on the left.
4. **No fractures:** There are no broken bones, but movement during the scan made some images harder to interpret.

General Spine Condition:

- The natural curve of your neck is straighter than usual, which can sometimes indicate muscle strain or alignment issues.
- The cushioning discs between your neck bones have lost some hydration, which is common with aging or degeneration.
- No major issues were found at the base of the skull or in the spinal cord itself.

Breakdown by Specific Levels:

- **C2-C3 (Upper Neck):** No significant issues.
- **C3-C4 (Upper-Mid Neck):** A mild disc bulge and minor joint wear but no significant narrowing.
- **C4-C5 (Mid-Neck):** More narrowing on the left side, with moderate-to-severe crowding of nerve spaces.
- **C5-C6 (Mid-Lower Neck):** Moderate narrowing in the center and in the nerve passageways on both sides.
- **C6-C7 (Lower Neck):** The worst narrowing is on the right side, where nerve spaces are significantly compressed.

What This Means for You:

- There is **moderate to severe narrowing in multiple areas**, which could contribute to nerve pain, tingling, or weakness in your arms.
- The loss of natural neck curvature might suggest **postural strain or muscle tightness**.
- No immediate surgical emergency is noted, but **nerve compression** may require treatment, depending on symptoms.

Current Theoretical Causes

My theories have been the following:

- Phone/Switch/Book usage while on the toilet (30ish minutes a day), which I've recently paid more attention to and realized I look down A LOT.
- Book reading in general, which I've ramped up on over the last few years.
- Driving, I tend to lead forward instead of keeping my head against the rest.
- Sleep, not having my neck in a correct position
- Last year as my eyes went bad, I was leaning forward a lot more until I got reading glasses, though I will tend to do this instead of making things easier to read, so I need to focus on that more.
- VR Headset: Because I was getting a strait neck in the past, this didn't cause the issues, but instead amplified them because my neck isn't properly curved to support the weight for long periods (an hour or so at a time).

Questions to ask about

Understanding the Cause & Progression

1. How long do you think these changes (disc osteophytes, narrowing, and loss of curve) have been developing?
2. Do my MRI findings suggest a natural (Non-aging) degenerative process, or could they be accelerated by posture, habits, or past injuries?
3. Is there any indication that a past injury (even one I don't recall) contributed to these issues?
4. Are these (non-aging) changes completely reversible with the right treatment, or is it more about managing symptoms and preventing further progression?

Symptoms & Daily Impact

1. Is the pain in my thumbs and hands likely connected to my neck issues, or is it a separate condition (such as tendonitis or nerve compression)?
 - a. Is it true that I am more susceptible to Repetitive Use Injuries because of this?
2. Are there warning signs I should look for that might indicate worsening nerve compression or more serious issues?

Treatment & Prevention (In addition to care plan below)

1. Are there any treatments or interventions (like injections, medications, or surgery) I should consider now or in the future?

Lifestyle & Long-Term Outlook

1. How do I know if my condition is improving or getting worse over time?
2. Would periodic follow-up imaging (like another MRI in a year or two) be necessary to monitor changes?

Lower Back Questions

1. I have yearly(**ish**) spasms in my lower back, which has also gotten worse in the last 6 months. X-rayed as a teenager, problems never discovered.
 - a. Is it likely these are from similar problems?
 - b. Is there any additional care planning which I should consider around the whole back instead of just the neck?

General

1. How can I know if I'm in the correct posture for my spine?

Current Theoretical Care Plan (To Modify during discussion)

Muscle Strengthening/Rebalancing

- I am currently working with a Muscle System Specialist (Physician's Fitness in Dublin)
 - He has already discovered and we are working on misalignments and weaknesses in my back and shoulders. Symptoms I believe have decreased from this.
 - Hoping this takes care of VR Headset issues long term

Pillow / Sleep on Back

- I have already been working forcing myself to sleep on my back.
- I bought a Cervical pillow to try that out while sleeping to help reduce my side to side head movement while sleeping
- Should I consider using a rolled towel/no pillow instead?

Driving

- I have to force myself to notice when my head is not touching the head rest.

Computer Usage

- I have a decently ergonomic setup at home, but have to focus on keeping my head back against the chair.
- I need to pay attention when I'm leaning forward to read, and use my reading glasses instead.
- Working in general is not as nicely setup, might need to make adjustments at the workplace.

Books/Phone Usage

- Pay attention? Hold up?

Cervical Traction

- Light at home traction, 5-10 Minutes, once a day. My understanding of benefits:
 - Creates **space between vertebrae**
 - Reduces **pressure on pinched nerves**
 - Helps **rehydrate discs** and improve mobility
 - Decreases **muscle tension and spasms**

Primary Strengthening

-

Diagnosis #3: Tech Neck

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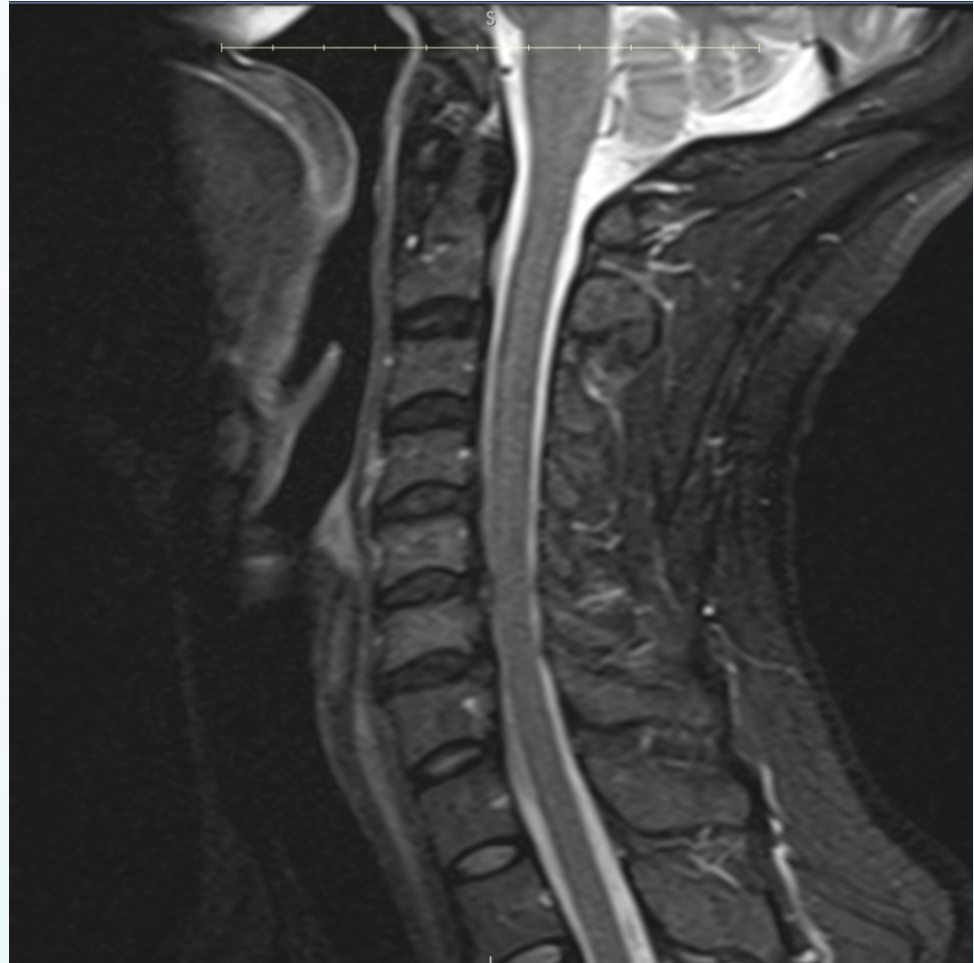
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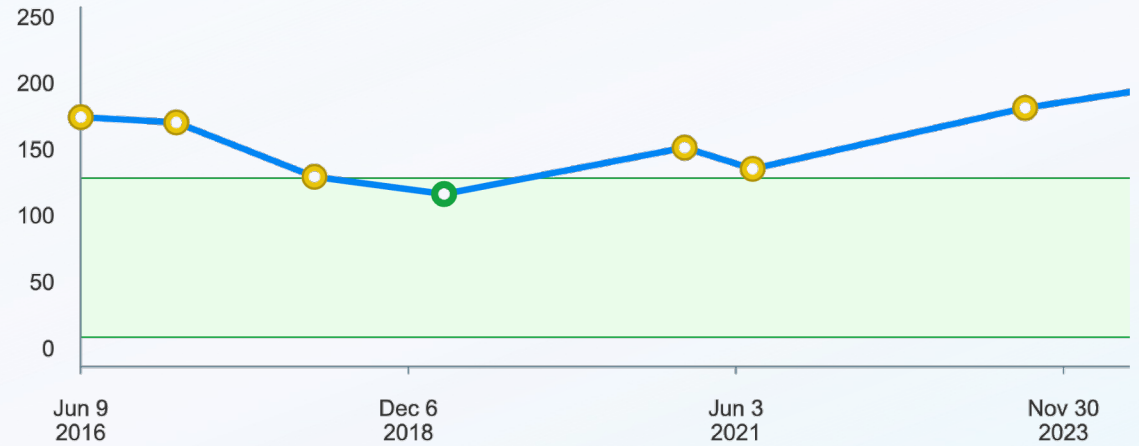
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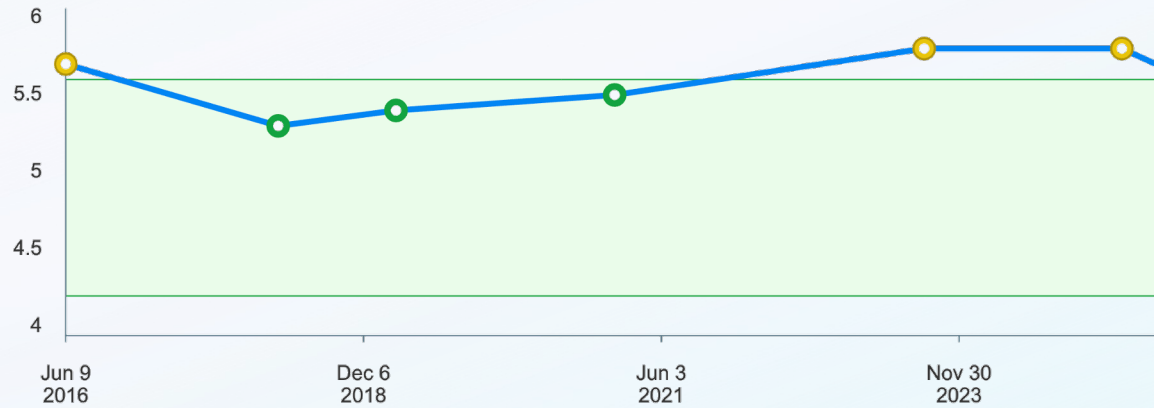


And The Metrics...

LDL Calculated
mg/dL



Hemoglobin A1C
Normal value: <5.7 %



Time for a real education....

PHYSICIANS FITNESS

MISSION STATEMENT

TO HELP INDIVIDUALS ACHIEVE INDEPENDENCE AND FREEDOM
SO THAT THEY CAN LIVE A HAPPY, SATISFYING, AND PAIN-FREE
LIFE BY BUILDING WELLNESS TOGETHER THROUGH FITNESS
EDUCATION, EXERCISE, AND MUSCLE/JOINT INJURY PREVENTION.

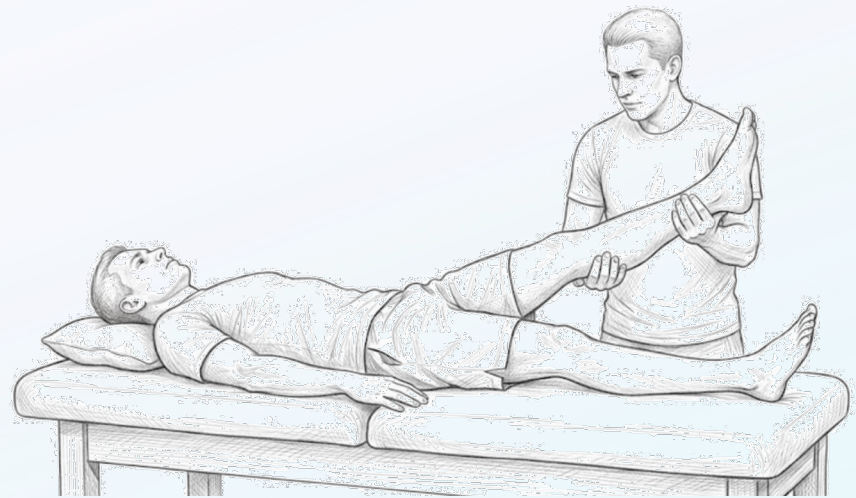
Assess Everything – Activate Everything

Muscle Activation

- Turn on the nervous system
- Connect your brain with the muscle

If I started here

- None of the above PT would be needed
- I also wouldn't know the root cause / complete state of my body



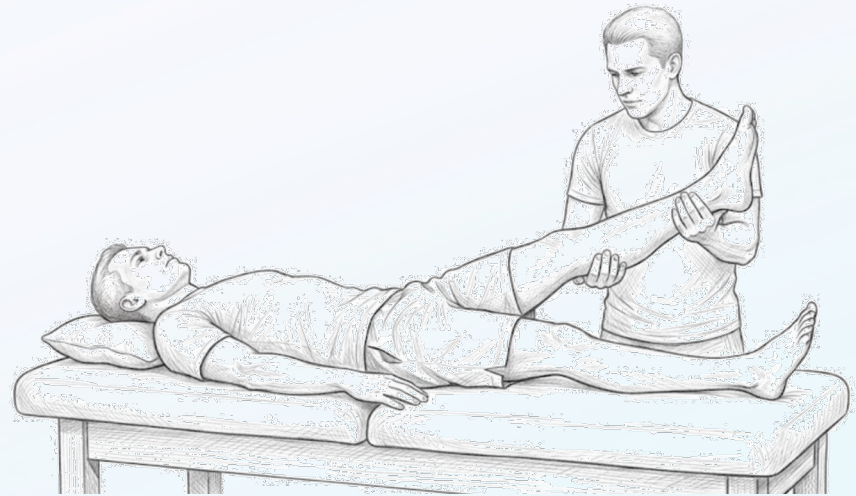
Immediate shifts – finally getting rid of the pain

What they noticed:

- I leaned heavy on my right foot
- I stood twisted
- I didn't walk correctly....

Primary focus areas:

- Back
- Knees
- Shoulders
- Ankles
- Glutes



More and Better Metrics



Metabolic Health

- Insulin Resistance
- LDL-P
- Apolipoprotein B
- Body Fat %
- Ketones

Physical Health

- Pain
- Grip Strength

My Health Manifesto

Long Term Habits over Short Term Goals

Natural Solutions over Medication and Drugs

End Measurements over Correlations

Minimal Effort over Perfection

Prevention over Reaction

Details – The Research

Physical Health: Muscle

Activation

- Building Habits to use Unused / Weak Muscles – Connect the Brain
- Focus Balancing the Body and Reducing Pain
- Helps Build correct Posture / Form

Lean

- Build Strength
 - Think “thicker” not “larger”
- Protein Intake shift*

Myth: Do NOT push to failure.

Bulk / Growth

- Build Strength
 - “larger” and “thicker”
- Major Protein Intake Shift*
- Creatine / Major Water Intake

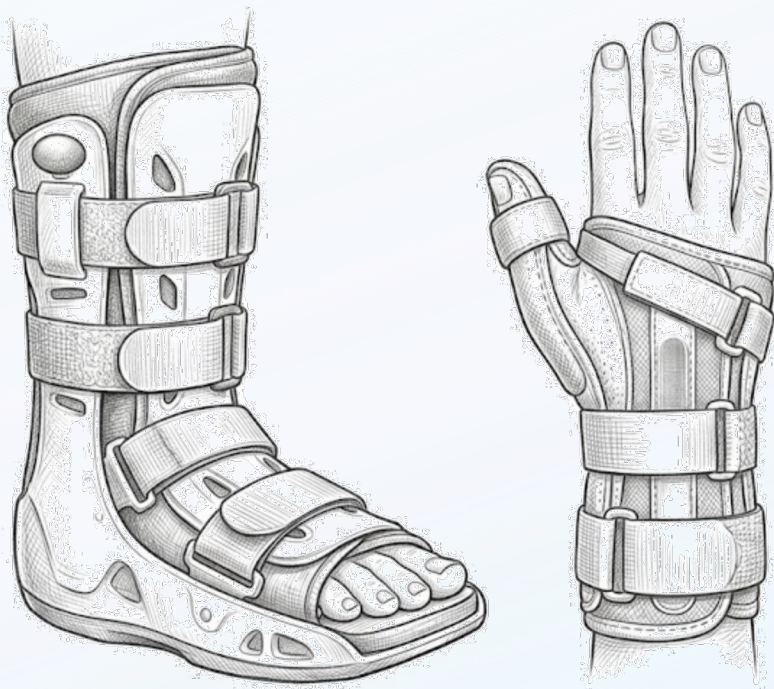
Physical Health: Activation

Quick Activation Checks

- When Gripping what Muscles do you feel?
- When Walking, what Muscles do you feel?

- Don't assume the cause and pain are in the same location
- Just about everything can start with the back / spine
- Check your posture regularly, bad habits will only make things worse over time
- Needs are based on your own actions: Work, Hobbies, past injuries.

Physical Health: Splints & Support



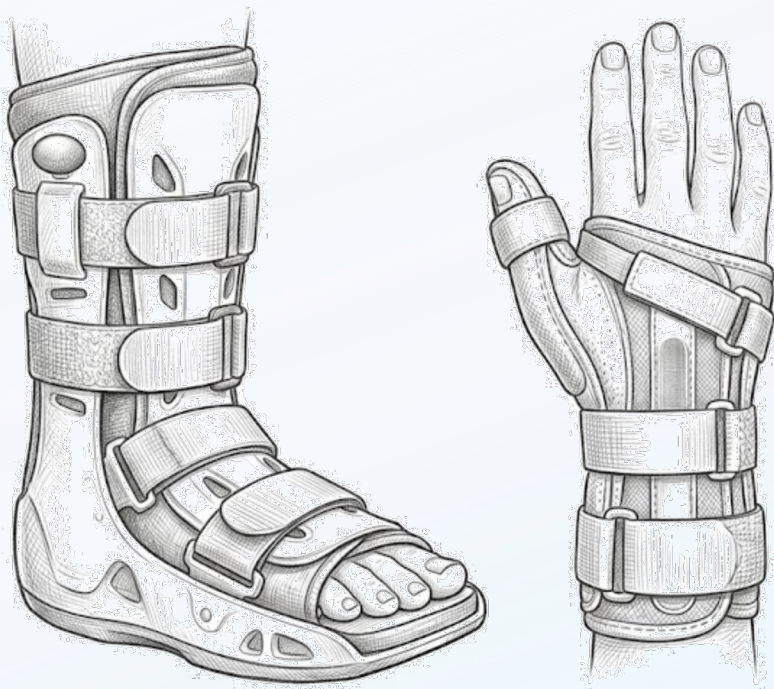
Hazards:

- Muscle deterioration
- Migrates muscle usages to other joints, in inappropriate ways
- Cause inappropriate postures

Prevention:

- Protein Intake
- Use when sleeping
- Physical Therapy

Physical Health: Splints & Support



Supports include:

- Lubar Supports
- Shoe Supports
- Running Shoes

Prevention:

- Protein Intake
- Use when sleeping
- Physical Therapy

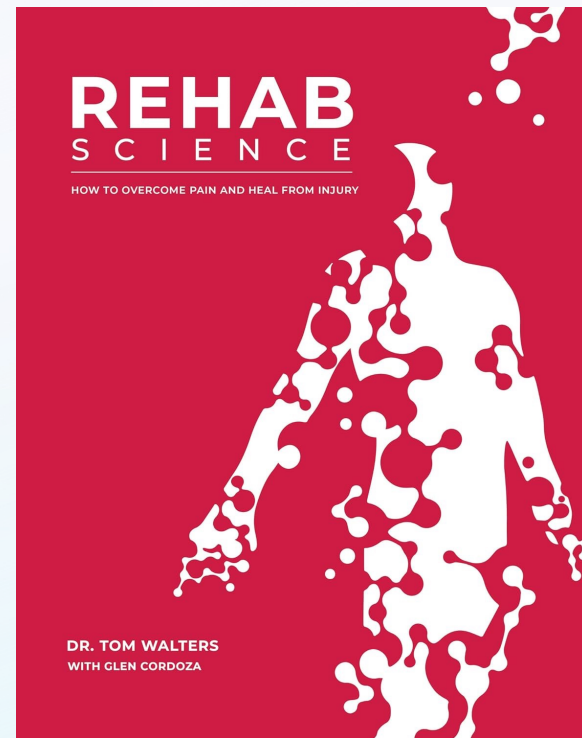
Physical Health: Physical Therapy

- You may not need Doctor approval for PT
- Ask what types of services they provide

Self Care – Use Discernment

- AI Can Help discover ways to test and exercises to try
- Many PT Videos

You do not need equipment



Protein

Standard Recognized Formulas

Convert weight to kg

Maintain: $\text{kg} * .7 = \text{g}$

Growth

- Min: $\text{kg} * 1.2 = \text{g}$

- Max: $\text{kg} * 2 = \text{g}$

Beware of your personal sensitivities
and medical conditions



**Yes, you CAN get your
protein in one meal**

Fiber

Standard(ish)

14g per 1000 Calories

- Soluble: 25%

Fermented Foods often
lacking in our diets



The Expert Quote

“The diet with *enough protein*, enough total calories, that is comprised predominately of **healthy food** choices that *fits the individual's personal preferences and tolerances*”

- Alan Aragon

Ketones & Insulin

The liver produces ketones when in a low-insulin state

Natural Ketones come from:

- Keto Based Diet
- Fasting (14 hour cycle, last to first meal)
 - Permits Black Coffee/Stright Tea, and water during fasting period

A1C != Insulin Resistance

“Insulin will tell every cell in the body from brain cells to bone cells, liver cells to lung cells, what it needs to do with energy...”

- Dr. Benjamin Bikman

Ketones are easy to measure

Psyllium Husk – Spread out Insulin Spikes

Chart Provided by "The Stress Paradox" - Sharon Horesh Bergquist

Marker	At risk	Room to improve	Optimal	Peak
Insulin resistance				
Fasting insulin level	≥15 mIU/L	7 to 14 mIU/L	3 to 6 mIU/L	<3 mIU/L
HOMA-IR	2.0 or higher	1.5 to 1.9	1 to 1.4	<1.0
Triglyceride/ HDL ratio	>2	1.5 to 2	1 to 1.4	<1.0

2023: $118/40 = \underline{\underline{2.95}}$

2024: $116/47 = \underline{\underline{2.46}}$

2025: $106/42 = \underline{\underline{2.52}}$

2026: $62/50 = \underline{\underline{1.24}}$

Cardio & HIITs

Reduces LDL

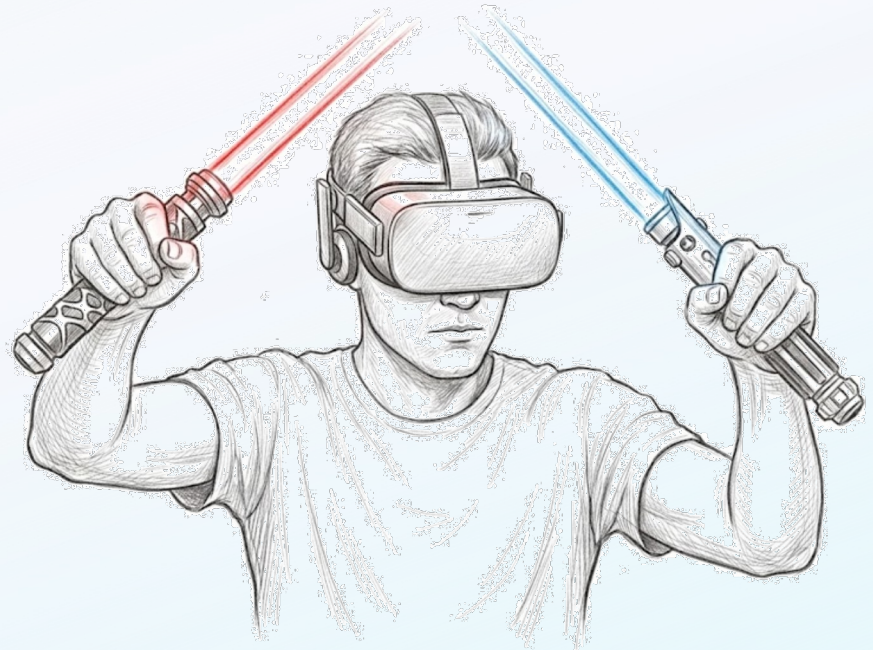
Raises HDL

Lowers Triglycerides

Long Cardio of Zone 3 and Higher are showing higher Coronary Artery Calcium scores*

HIITs – Better Heart Health

- Zone 2: ~15-20 Minutes
- Zone 5 : 30-60 seconds bursts



* <https://pmc.ncbi.nlm.nih.gov/articles/PMC13084289/>

Motivation / Willpower

Minimums, NO Maximums

No Streak Tracking – Build Currency

Communicate Plan and Goals to Friends

Do not eliminate anything

Delayed Gratification

Make the choice ahead of time

Mini / Tiny / Atomic Habits

My Strategies – Putting it all Together

Successes, and Failures

Body Fat %: 20% -> 15%

Grip Strength: 90lbs -> 130lbs

Insulin Resistance: At Risk -> Optimal

Numb Fingers:

- Daily to Never
- VR for extended periods

Pain:

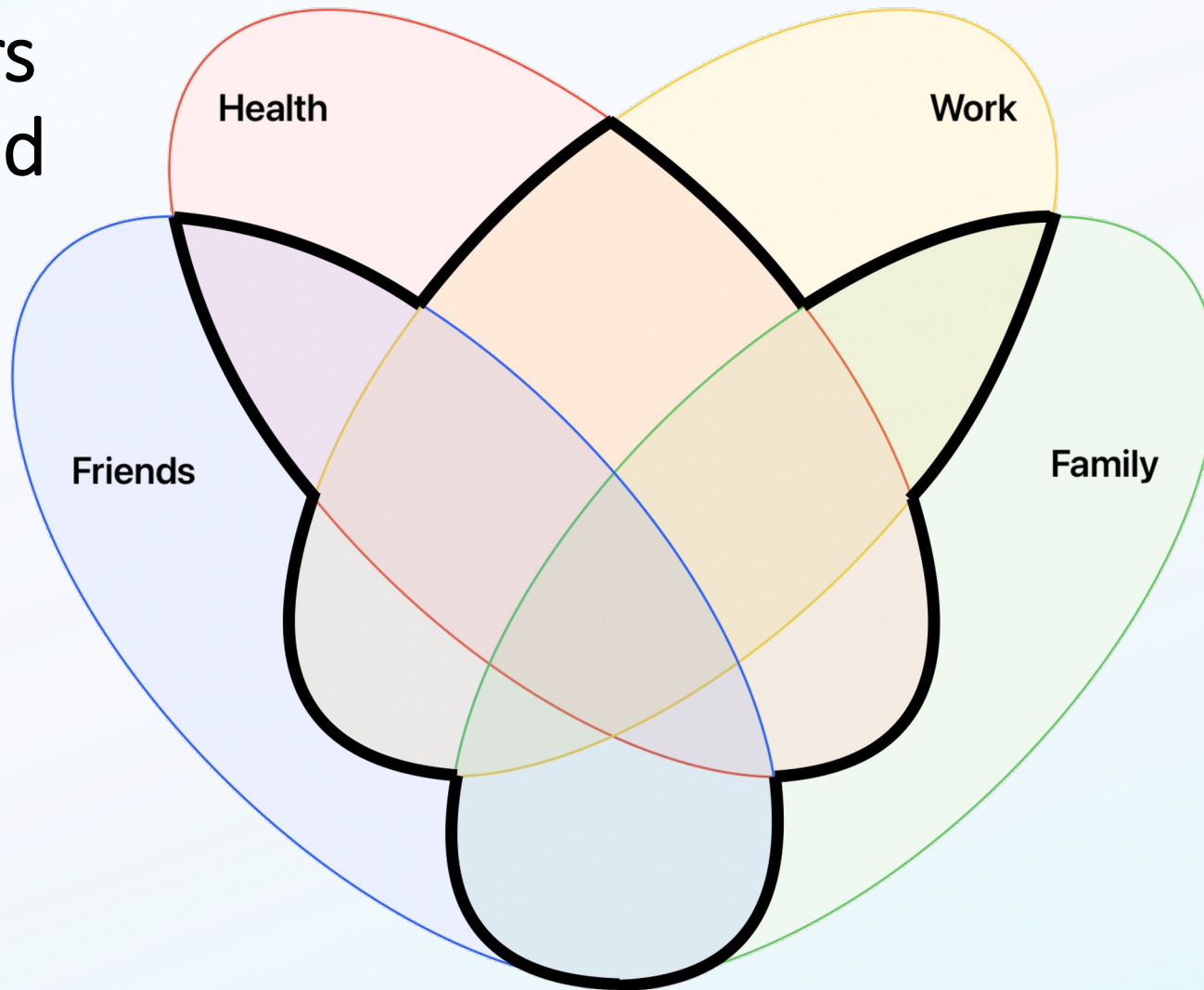
- Walk for several miles with none
- Back Spasm Pain reduced 90%

LDLs, Apolipoprotein B, LDL-P: Bad to Worse

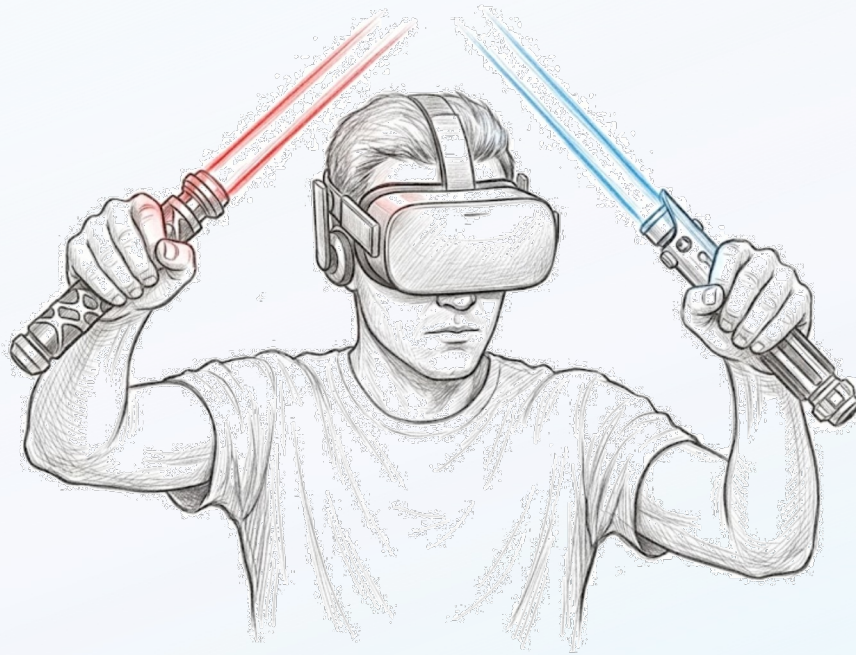
Sleep – Still figuring it out

- Goal: Keep Spine Strait

4 Burners Reworked



Cardio – Know One-Self - Multiples



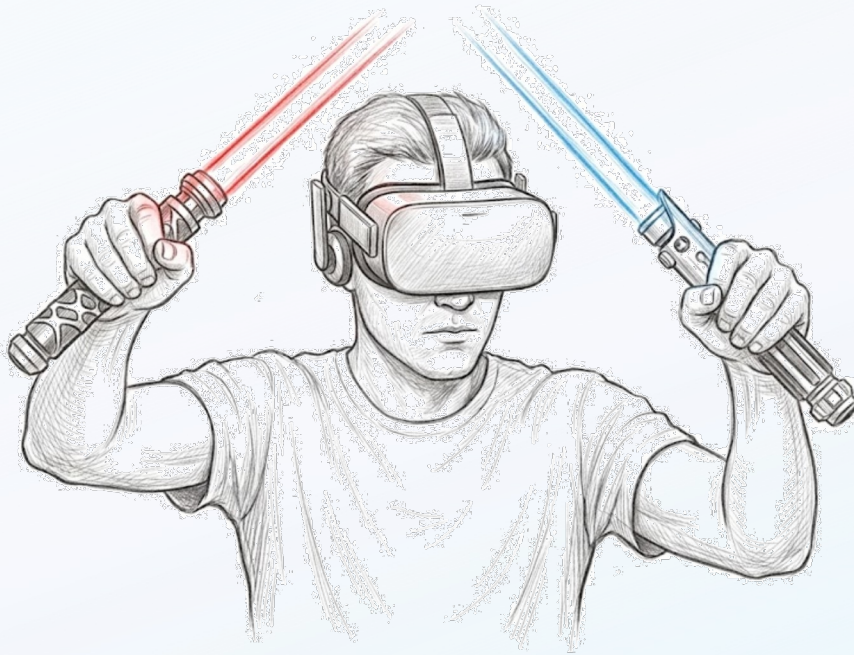
Home only

One-time Purchases

Multiple Strategies

- Exercise Bike
- Rower
- VR
- Weighted Vest

Muscle Activation / Lean Muscle Growth



Home only

One-time Purchases

Adjustable Weights

PT Bands

Hand Gripper / Finger Bands

^ Put at Desk

Food Lifestyle – “Dirty” Keto – Reduce Liver Pressure

Minimums, NO Maximums

- 70g Protein
- 25g Fiber

Groceries MUST go towards those goals

Avoid drinking sugar

14 Hour Fast: ~10pm to Noon

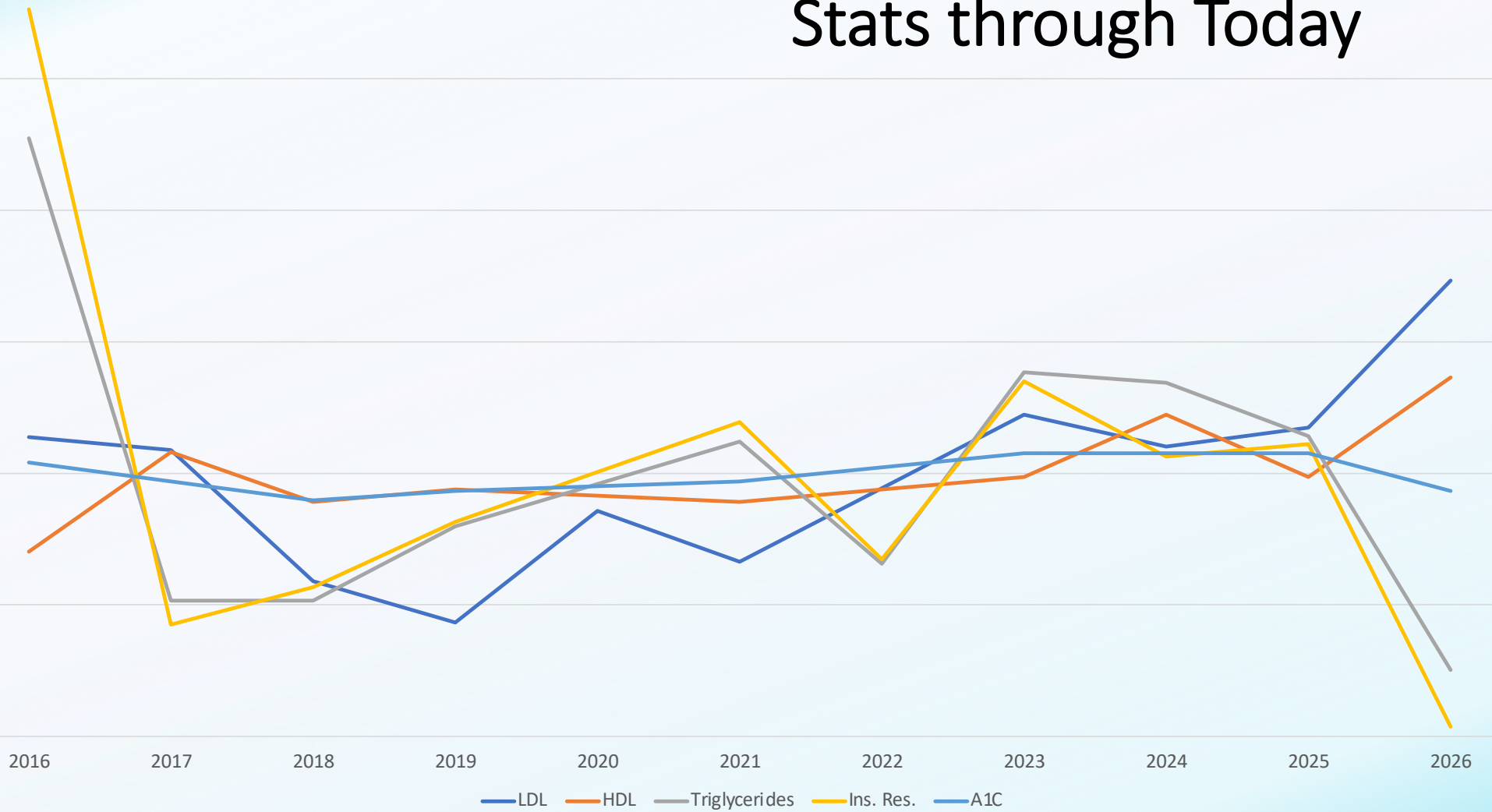


Supplement: Psyllium Husk

Medication: Statin

Supplements: D, Multi

Stats through Today



And then what?

Where do you go from here?

What does your health journey look like?

What measurements do you care about?

- What are your current measurements?
- What are your goal measurements that will lead you to a life you want to live?

What exercises do you enjoy doing, or would like to try?

What are your healthy personal food tolerances?

- What foods do you love?
- What foods can you not tolerate?

How can you integrate your health plans with your friends, family, and/or work?

- Name one friend, right now, who you trust and will follow up with

What does your health journey look like?

What Habits have you successfully built?

What habits have you not?

What means you can get Physical Therapy assistance made for you?

Do you have a regular doctor?

- If not, what is your next step to getting one?